# Shrishti Tomar

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## Summary

Computer Science undergraduate (B.Tech, 2027) building backend and full-stack systems with Node.js, TypeScript, PostgreSQL, Redis, and Docker. Shipped four live projects, including a job queue engine and RAG code search.

## Education

**Guru Gobind Singh Indraprastha University** — Delhi, India
B.Tech, Computer Science and Engineering; CGPA: 8.29 / 10 — *Expected June 2027*
Class XII; Percentage: 94.6%
**Relevant Coursework:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## Projects

**RepoInsight** | *AI Code Search Platform (RAG)* — GitHub | Live

- Built a **Retrieval-Augmented Generation (RAG)** platform that chunks GitHub repos and indexes them as **Gemini vector embeddings** in **MongoDB Atlas Vector Search** for natural-language code Q&A.
- Designed an agentic **LangGraph** workflow (retrieve, generate, tool calls, self-evaluate) on **Groq**-hosted LLMs, combining vector and keyword retrieval to ground every answer in cited source files.
- Containerized the React/Vite frontend and Express backend into a single **Docker** image on **AWS EC2**, with **GitHub OAuth**, **JWT** authentication, login rate limiting, and chat history in MongoDB Atlas.

**TaskForge** | *Distributed Task Queue Engine* — GitHub | Live

- Engineered an asynchronous job engine in **TypeScript** with **Express, BullMQ, and Redis**, offloading OCR (Tesseract), scraping, PDF reports, CSV exports, and signed webhooks to background workers.
- Load-tested the API at **310 req/s with 60 ms p50 latency** while 50 OCR jobs ran (**3 jobs/s**); streamed live job progress through Redis pub/sub and **Socket.IO** WebSockets, replacing HTTP polling.
- Added exponential-backoff retries, a **dead-letter queue**, **idempotency keys**, and an **SSRF guard**; secured the API with dual-token **JWT** auth and **RBAC** over **PostgreSQL** schemas modeled in **Prisma ORM**.
- Wrote **70 Jest unit and integration tests** run in a **GitHub Actions CI** pipeline against real PostgreSQL and Redis; orchestrated a 5-container **Docker Compose** stack.

**PrepWise** | *AI Career Coach and Mock Interview Platform* — GitHub | Live

- Built a full-stack AI career-prep SaaS with **Next.js** (App Router, Server Actions) where the **Gemini API** generates role-specific mock interview quizzes and personalized feedback, tracked on a progress dashboard.
- Added AI-assisted resume and cover-letter builders with PDF export, weekly industry-insight generation via **Inngest** cron jobs, **Clerk** authentication, and **PostgreSQL** with **Prisma ORM**.

**DropAlert** | *E-Commerce Price Tracker* — GitHub | Live

- Built a **Next.js** and **Supabase** price tracker that extracts structured product data (name, price, currency, image) from JavaScript-heavy store pages with the **Firecrawl API**, plus Google OAuth via Supabase Auth.
- Automated price checks through a secret-protected **cron** API route, caching scrape results in **Redis** (1-hour TTL) and rate-limiting product additions per user to cut redundant API calls.
- Sent templated HTML email alerts via **Resend** on price drops; charted price history with Recharts.

## Technical Skills

**Languages**: C++, Java, JavaScript, TypeScript, Python, SQL
**Frontend**: React, Next.js (App Router, SSR), Redux Toolkit, TanStack Query, Tailwind CSS
**Backend**: Node.js, Express.js, REST APIs, BullMQ, WebSockets (Socket.IO), JWT, OAuth, RBAC
**Databases**: PostgreSQL, MongoDB, Redis, Supabase, Prisma ORM
**AI / LLM**: RAG, LangChain, LangGraph, Vector Embeddings, Vector Search, Gemini API, Groq
**Cloud and DevOps**: AWS EC2, Docker, Docker Compose, CI/CD (GitHub Actions), Linux, Git, Vercel, Render
**Testing**: Jest, Unit and Integration Testing, Load Testing

## Achievements

- **GTBIT Hackathon:** Secured 3rd place among 80+ competing teams.
- **HackMait Hackathon:** Placed in the top 17 teams out of 700+ participants.
- **Competitive Programming:** Solved 400+ DSA problems on LeetCode and other platforms.